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Annexe 2
Appendix 2

Recommended Values of Standard Frequencies for Applications Including the Practical Realization of the Metre and Secondary Representations of the Definition of the Second: Rubidium ($\lambda \approx 780\ \mathrm{nm}$)

May 2016
mai 2016

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^{87}Rb atom, crossover transition between the d and f hyperfine components of the saturated absorption at 780 nm **(D2 transition)**

1. Recommended value [1] of the frequency:

$f\left({}^{87}\mathrm{Rb}_{\mathrm{d/f\ crossover}}\right) = 384\,227\,981.9\ \mathrm{MHz}$

equivalent to

$\lambda\left({}^{87}\mathrm{Rb}_{\mathrm{d/f\ crossover}}\right) = 780\,246\,291.6\ \mathrm{fm},$

with an estimated relative standard uncertainty of 5×10^{-10} applies to the radiation of a tunable Extended Cavity Diode Laser, stabilized to the d/f saturated absorption line in a well characterized magnetically shielded rubidium absorption cell by the third-harmonic technique,

- cell temperature (22.5 ± 1) °C
- frequency modulation width, peak-to-peak, (5 ± 1) MHz;
- saturating beam intensity of (120 ± 20) mW cm^{-2}

2. Source data

Adopted value $f = 384\,227\,981.9$ MHz

$u_c/y = 5 \times 10^{-10}$

calculated from

$f(^{87}\mathrm{Rb}_{\mathrm{d/f\ crossover}})$/kHz	u_c/y	source data
384 227 981 877.3(5.5)	1.5×10^{-11}	[2]
384 227 981 867.8(165)	4.3×10^{-10}	[3]

as a weighted mean. The fractional uncertainty was evaluated to be approximately 4.3×10^{-10} for a 0.01 s averaging time [3]. For the recommendation this uncertainty was rounded to 5×10^{-10}.

References

[1] CIPM Recommendation 2 (CI-2015): Updates to the list of standard frequencies http://www.bipm.org/jsp/en/CIPMRecommendations.jsp

[2] J. Ye, S. Swartz, P. Jungner, and J. L. Hall, "Hyperfine structure and absolute frequency of the ^{87}Rb $5P_{3/2}$ state", *Opt. Lett.* **21**, 1280 (1996).

[3] Y. Bitou, K. Sasaki, H. Inaba, F.-L. Hong, A. Onae, "Rubidium-stabilized diode laser for high-precision interferometer", *Opt. Eng.* **43**, 900 (2004).